

 Complete Presentation Script (15 Slides)

Advanced Arithmetic Optimization Algorithm (nAOA)

Agushaka & Ezugwu (2021) - PLOS ONE

 Presentation Overview

Item	Details
Total Slides	15
Duration	25-30 minutes
Audience	Academic/Research, Graduate students, Engineers

Slide Roadmap:

#	Topic	Time
1	Title	0:30
2	Engineering Design Problems	2:00
3	Traditional Methods & Limitations	1:30
4	The Research Gap	2:00
5	Foundation: Metaheuristics	1:30
6	Evolution: SCA to AOA	1:30
7	How nAOA Works	2:00
8	Exploration vs Exploitation	2:00
9	Results: Benchmarks	2:00
10	Results: Engineering Problems	2:30
11	Results: Convergence	1:30
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13	Conclusion	1:30
14	Reflection: Strengths & Limitations	2:00
15	Application to Our Project	2:00

SLIDE 1: Title Slide

 Script (30 seconds)

"Good [morning/afternoon] everyone. Today I'll present the literature review for 'Advanced Arithmetic Optimization Algorithm for Solving Mechanical Engineering Design Problems.'

This research by Agushaka and Ezugwu, published in PLOS ONE 2021, introduces nAOA – a significant improvement to the original Arithmetic Optimization Algorithm. Let's explore what problems this addresses and how it works."

 Key Points

- Establish credibility: peer-reviewed journal
- Preview structure: problem → solution → results → application

SLIDE 2: Engineering Design Problems

 Script (2 minutes)

"Before diving into algorithms, let's understand the PROBLEMS we're trying to solve.

Welded Beam Design (WBD) – Minimize manufacturing cost with 4 design variables (weld thickness, length, height, beam thickness) and 7 constraints for stress, deflection, and buckling.

Compression Spring Design (CSD) – Minimize spring volume. 3 variables: wire diameter, coil diameter, number of active coils. 4 constraints.

Pressure Vessel Design (PVD) – Minimize total cost of a cylindrical vessel. 4 variables controlling thicknesses, radius, and length.

What makes these difficult? They're **constrained, nonlinear, and multimodal** – traditional calculus methods often fail."

 INSERT GRAPHICS

- **Figure 6** (page 19): Welded Beam diagram
- **Figure 8** (page 22): Compression Spring diagram
- **Figure 10** (page 24): Pressure Vessel diagram

SLIDE 3: Traditional Methods & Limitations

 Script (1.5 minutes)

"Why can't we use traditional optimization methods?

Traditional methods include gradient-based approaches (Newton's method, conjugate gradient), linear programming (Simplex), and dynamic programming.

Why they fail for engineering problems:

1. **Require gradient information** – fails with non-differentiable functions
2. **Get stuck in local optima** – miss the global best
3. **Sensitive to starting point** – inconsistent results
4. **Struggle with constraints** – engineering has many inequality constraints

This is why metaheuristic algorithms emerged."

💬 Engagement

"Has anyone experienced gradient-based optimizers getting stuck?" (pause for audience)

SLIDE 4: The Research Gap

🖋️ Script (2 minutes)

"Here's the heart of this paper – the research gap.

The original AOA uses basic operators: **multiply, divide, add, subtract**. While effective, they have **limited exploration capability** – they make incremental changes, potentially missing promising regions.

nAOA's solution: Introduce **natural logarithm (ln)** and **exponential (e^x)** operators.

Why? Log compresses large ranges; exponential amplifies small differences. This creates **high dispersion values** for effective exploration.

Addition and subtraction are kept for exploitation (refinement).

Result: **Enhanced exploration AND exploitation** – the best of both worlds."

📌 This is the CORE slide

- Gap: limited exploration in original AOA
- Solution: ln and exp operators

SLIDE 5: Foundation - Metaheuristic Algorithms

🖋️ Script (1.5 minutes)

"Why metaheuristics are better for these problems:

✓ **Gradient-free** – no derivatives needed ✓ **Bypass local optima** – explore broadly before converging ✓ **Easy to implement** – few dozen lines of code ✓ **Domain-agnostic** – same algorithm works across applications

Categories:

- **Bio-inspired:** GA, AAA, ES (mimic evolution)

- **Physics-based:** SA, GSA, ACROA (mimic physical laws)
- **Swarm-based:** PSO, ACO (mimic collective behavior)

AOA and nAOA are unique – purely mathematical, not nature-inspired."

SLIDE 6: Evolution Timeline

 Script (1.5 minutes)

"Evolution leading to nAOA:

2016: Sine Cosine Algorithm (SCA) – proved mathematical functions work for optimization

2021: AOA – used basic arithmetic operators

2021: nAOA (this paper) – enhanced with ln and exp operators

Key competitors compared:

- SCA, SSA, SMA, GWO

Important finding: GWO (Grey Wolf Optimizer) was the top performer in engineering problems. nAOA aims to match or exceed this."

SLIDE 7: How nAOA Works

 Script (2 minutes)

"The algorithm flow:

Step 1: Initialize – Generate solutions using **beta distribution** (not uniform random)

Step 2: Evaluate – Calculate fitness for each solution

Step 3: Decide Phase – Compare random β_1 with MOA (Math Optimization Accelerator)

Step 4: Update – If exploring: use ln/exp. If exploiting: use +/-

Step 5: Iterate – Repeat until max iterations

Key equations:

- **MOA** controls exploration vs exploitation balance (increases over iterations)
- **MOP** controls step size (decreases over iterations)"

 INSERT GRAPHIC

- **Figure 4** (page 8): nAOA Flowchart

SLIDE 8: Exploration vs Exploitation

 Script (2 minutes)

"Two phases in detail:

EXPLORATION (when $\beta_1 < \text{MOA}$):

- In operator (when $\beta_2 < 0.5$)
- exp operator (when $\beta_2 \geq 0.5$)
- Purpose: Search NEW regions with high dispersion

EXPLOITATION (when $\beta_1 > \text{MOA}$):

- Subtraction (when $\beta_3 < 0.5$)
- Addition (when $\beta_3 \geq 0.5$)
- Purpose: Refine near best solution with low dispersion

Key insight: Early iterations favor exploration (low MOA), later favor exploitation (high MOA). This mimics human problem-solving – broad search first, then refinement."

 INSERT GRAPHICS

- **Figure 1** (page 4): Effect of operators
- **Figure 2** (page 5): Exploration/exploitation phases

SLIDE 9: Results - Benchmark Functions

 Script (2 minutes)

"Testing on **30 benchmark functions**:

- **Unimodal (F1-F7):** Tests exploitation – 7 functions
- **Multimodal (F8-F20):** Tests exploration – 13 functions
- **CEC 2020 (F21-F30):** Tests escaping local minima – 10 functions

Results (Friedman test, $p=0.00$):

- Unimodal: nAOA ranked **#1** (mean rank: 3.48)
- Multimodal: nAOA ranked **#1**
- CEC 2020: nAOA ranked **#1** (mean rank: 2.9)

nAOA outperformed: AOA, CPSOGSA, GSA, PSO, BBO, DE, ACO, SSA, SCA"

 INSERT GRAPHIC

- **Figure 5** (page 18): Selected convergence curves (F1, F9, F21)

SLIDE 10: Results - Engineering Problems

 Script (2.5 minutes)

"Real engineering results:

Problem	Best Cost	nAOA Avg	AOA Avg	GWO Avg	Rank
WBD	1.6976	1.7731	2.3488	1.6976	2nd
CSD	3.6619	3.6819	6.1167	3.6619	2nd
PVD	2302.5	3303.1	4440.8	2556.8	2nd

Key findings:

- 1. **nAOA vs AOA:** 24-44% better average costs – substantial improvement
- 2. **GWO comparison:** GWO was best overall, but nAOA matched GWO's best values in WBD and CSD"

SLIDE 11: Results - Convergence Behavior

 Script (1.5 minutes)

"Convergence analysis:

Four key observations:

- 1. nAOA converges **smoothly without premature convergence**
- 2. **Better exploration** in early iterations (higher diversity)
- 3. **Effective exploitation** in later iterations (rapid convergence)
- 4. **Insensitive to initialization** (beta distribution helps)"

 INSERT GRAPHICS (CRITICAL)

- **Figure 7** (page 21): WBD convergence curves
- **Figure 9** (page 23): CSD convergence curves

SLIDE 12: Statistical Validation

 Script (1.5 minutes)

"Statistical tests for credibility:

Wilcoxon signed-rank test:

- Significant ($p < 0.05$): SSA, SCA, GWO, nAOA, AOA, CPSOGSA, GSA

- Not significant: PSO, BBO, DE, ACO (poor performance)

Friedman ranking test ($p = 0.00$):

- Classical functions: nAOA #1 (rank 3.48)
- CEC 2020: nAOA #1 (rank 2.9)

Conclusion: Results are statistically significant, not random chance."

SLIDE 13: Conclusion

 Script (1.5 minutes)

"Four main conclusions:

1. **nAOA successfully improves AOA** – In/exp operators enhance exploration
2. **Beta distribution is effective** – better than uniform for initialization
3. **Competitive performance** – #1 in benchmarks, #2 in engineering
4. **Statistically validated** – Friedman and Wilcoxon tests confirm significance

Future directions: Chaotic maps, hybridization, more applications"

SLIDE 14: Reflection - Strengths & Limitations

 Script (2 minutes)

"Strengths:

- Simple implementation (4 operators)
- No complex parameter tuning ($\alpha=5$, $\mu=0.5$ works)
- Solid mathematical foundation
- Good exploration-exploitation balance

Limitations:

- Not always best (GWO wins in engineering)
- Limited test scope (only 3 engineering problems)
- Premature convergence risk (chaotic maps suggested)
- Computational cost $O(N \times (IP + 1))$

Critical question: Is the improvement over AOA worth the In/exp complexity? For most applications, yes."

SLIDE 15: Application to Our Project

 Script (2 minutes)

"When to use nAOA:

- Continuous optimization problems
- Constrained engineering design
- Multi-modal landscapes
- When simple implementation needed
- Benchmarking against other methods

Implementation steps:

1. Define objective function $f(x)$
2. Set bounds [LB, UB]
3. Initialize with beta distribution
4. Set $\alpha=5$, $\mu=0.5$, iterations=1000
5. Run nAOA, compare with baselines

[CUSTOMIZE FOR YOUR PROJECT:]

- Parameter optimization for ML models
- Mechanical/structural design optimization
- Resource allocation and scheduling"

 COMPLETE GRAPHICS CHECKLIST

Slide	Figure	Description	Paper Page
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 KEY NUMBERS TO REMEMBER

Metric	Value
Benchmark functions tested	30

Metric	Value
Engineering problems	3
Benchmark ranking	#1
Engineering ranking	#2 (behind GWO)
Friedman test p-value	0.00
Default α	5
Default μ	0.5

 nAOA vs AOA Quick Reference

Aspect	Original AOA	nAOA
Exploration operators	$\times, \div$	$\ln, e^x$
Exploitation operators	$+, -$	$+, -$
Initialization	Uniform	Beta distribution
Random variables	Uniform	Beta distributed
Benchmark rank	3rd-4th	1st
Engineering rank	3rd-4th	2nd

Script Complete | 15 Slides | 25-30 minutes